

TOPICAL PRACTICE — WEEK 1

Physical Chemistry — Topic 1: Atomic Structure

Monday 17 August 2026 — Friday 21 August 2026

CANDIDATE

Urwah

SUBJECT

**Chemistry — Cambridge
International A Level (9701)**

TOTAL MARKS (WEEK)

134 marks**HOW TO USE THIS PACK**

- One short set of questions per session, matching that day's topic from the schedule.
- Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].
- Mark schemes for the whole week are collected at the end of the pack.
- These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text (see the note on the final page).

THIS WEEK AT A GLANCE

- | | |
|--|---|
| ▶ Mon 17 Aug — 1.1 Particles in the Atom & Atomic Radius (22 marks, incl. graph) | ▶ Thu 20 Aug — 1.3 Electron Configuration & Orbitals, Part 2 (24 marks) |
| ▶ Tue 18 Aug — 1.2 Isotopes (20 marks, incl. graph) | ▶ Fri 21 Aug — Paper 5 Focus: Planning & Analysis (26 marks, incl. graph) |
| ▶ Wed 19 Aug — 1.3 Electron Configuration & Orbitals, Part 1 (22 marks, incl. graph) | ▶ Mixed — Full-Week Synthesis Question (20 marks, incl. graph) |

MONDAY · 17 AUG — TOPIC 1.1: PARTICLES IN THE ATOM & ATOMIC RADIUS
1 Multiple Choice

Syllabus 1.1

For each question, put a cross **X** in the box next to your chosen answer.

- (a) Which particle has a relative charge of +1 and is found in the nucleus? [1]
A electron **B** neutron
C proton **D** positron
- (b) An atom has proton number 15 and nucleon number 31. How many neutrons does it contain? [1]
A 15 **B** 16
C 31 **D** 46
- (c) When a beam of electrons passes through a uniform electric field, the electrons are deflected towards: [1]
A the positive plate **B** the negative plate
C neither plate **D** both plates equally
- (d) Which row shows the correct trend in atomic radius and first ionisation energy across Period 3 (Na to Ar)? [1]
A radius decreases, IE increases **B** radius increases, IE increases
C radius decreases, IE decreases **D** radius increases, IE decreases
- (e) Which ion is smaller than its parent atom? [1]
A Cl^- **B** O^{2-}
C Na^+ **D** S^{2-}
- (f) Down a group of the Periodic Table, atomic radius increases mainly because: [1]
A nuclear charge decreases **B** the number of occupied electron shells increases
C shielding decreases **D** the number of protons decreases

2 Sub-Atomic Particles

Syllabus 1.1

Complete the table to show the relative charge and relative mass of each sub-atomic particle.

Particle	Relative Charge	Relative Mass
Proton		
Neutron		
Electron		

[3]

3 Behaviour in an Electric Field

Syllabus 1.1

A beam containing protons, neutrons and electrons is passed through a uniform electric field created by two charged parallel plates.

- (a) Describe and explain what happens to the protons as they pass through the field. [2]
.....
.....
- (b) Explain why the neutrons are undeflected. [1]
.....

4 Interpreting a Trend Graph

Syllabus 1.1

The graph below shows how atomic radius varies across Period 3 of the Periodic Table.

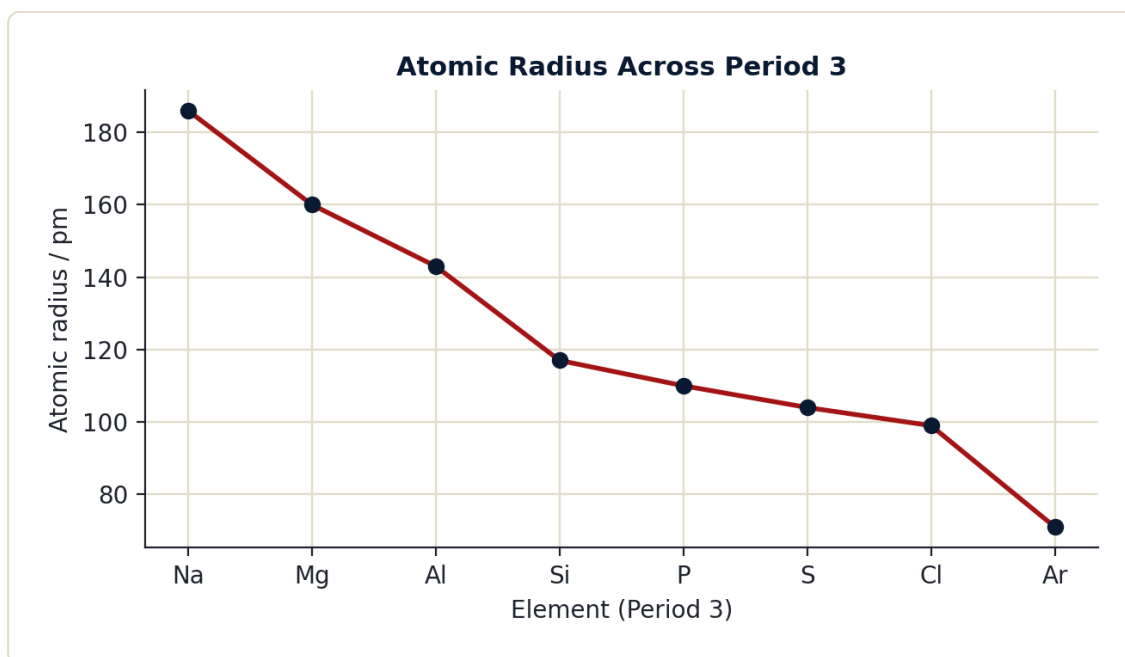


Fig. 4.1 Atomic radius across Period 3

- (a) Use the graph to state the atomic radius of silicon, and of chlorine. [2]
- (b) Describe the overall trend shown by the graph. [1]
- (c) Explain this trend in terms of nuclear charge and shielding. [3]

5 Atomic vs Ionic Radius

Syllabus 1.1

- (a) Explain why the radius of a Na^+ ion is smaller than the radius of a Na atom. [2]
- (b) Explain why the radius of a Cl^- ion is larger than the radius of a Cl atom. [2]

TUESDAY · 18 AUG — TOPIC 1.2: ISOTOPES
6 Multiple Choice

Syllabus 1.2

- (a) Isotopes of the same element have: [1]
A the same number of protons and neutrons **B** the same number of protons, different neutrons
C the same number of neutrons, different protons **D** different numbers of protons and electrons
- (b) Which property is different between two isotopes of the same element? [1]
A number of electron shells **B** chemical reactivity
C density of the pure element **D** number of outer-shell electrons
- (c) In a mass spectrometer, ions are separated according to their: [1]
A boiling point **B** mass-to-charge ratio (m/z)
C electronegativity **D** ionisation energy
- (d) An element has two isotopes: 90% at mass 20 and 10% at mass 22. Its relative atomic mass is closest to: [1]
A 20.0 **B** 20.2
C 21.0 **D** 22.0

7 Defining & Representing Isotopes

Syllabus 1.2

- (a) Define the term *isotope*. [2]
.....
.....
- (b) Chlorine exists as two isotopes. Using the correct notation, write the full isotopic symbol for chlorine-35 and chlorine-37. [2]
.....
.....

8 Chemical vs Physical Properties

Syllabus 1.2

- (a) Explain why chlorine-35 and chlorine-37 have identical chemical properties. [2]
.....
.....
- (b) Suggest **one** physical property that would differ slightly between the two isotopes, and explain why. [2]
.....
.....

9 Interpreting a Mass Spectrum

Syllabus 1.2 / 2.1

The mass spectrum of a sample of Element Q is shown below.

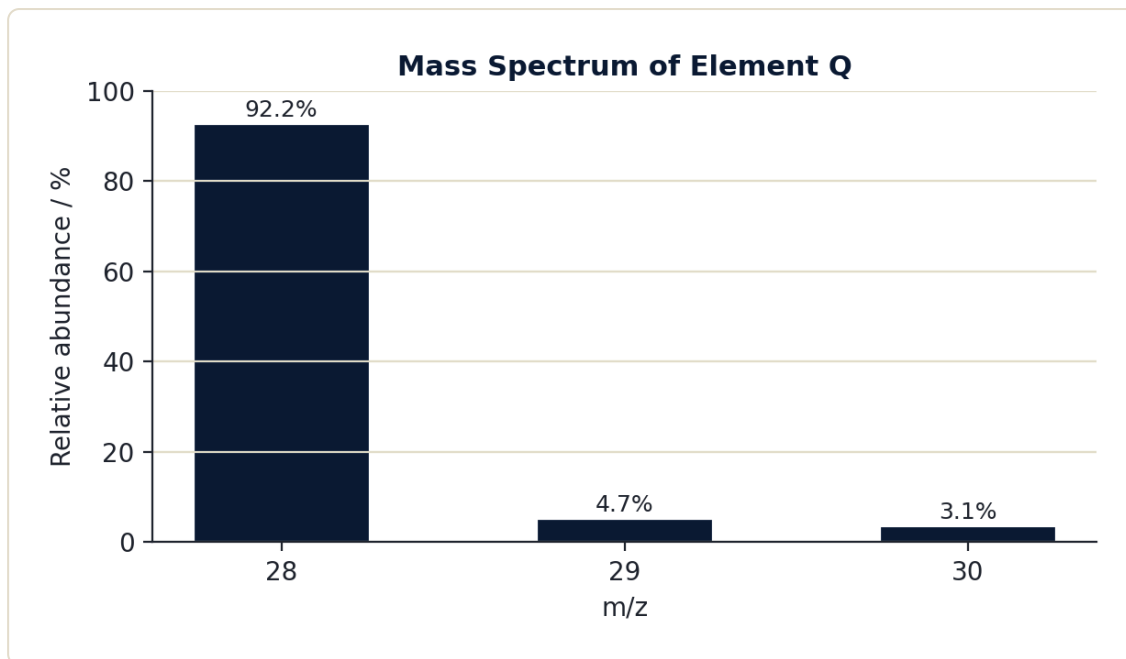


Fig. 9.1 Mass spectrum of Element Q

- (a) State the number of isotopes of Element Q shown in the spectrum. [1]
-
- (b) Calculate the relative atomic mass of Element Q, using the data from the graph. Give your answer to 1 decimal place, showing your working. [3]
-
-
-
- (c) Suggest why the peak at $m/z = 29$ is much smaller than the peak at $m/z = 28$. [1]
-

10 Applications of Isotopes

Syllabus 1.2 (extension)

- (a) Suggest **one** real-world application of isotopes (e.g. in medicine, dating, or industry), and briefly explain the property of isotopes that makes this application possible. [3]
-
-
-

WEDNESDAY · 19 AUG — TOPIC 1.3: ELECTRON CONFIGURATION & ORBITALS (PART 1)

11 Multiple Choice

Syllabus 1.3

- (a) The maximum number of electrons that can occupy a *p* sub-shell is: [1]
A 2 **B** 6
C 8 **D** 10
- (b) How many orbitals are there in a *d* sub-shell? [1]
A 1 **B** 3
C 5 **D** 7
- (c) Which sub-shell is normally filled immediately **before** the 3d sub-shell? [1]
A 3p **B** 3s
C 4s **D** 4p
- (d) The shape of an s orbital is best described as: [1]
A dumb-bell **B** spherical
C planar **D** tetrahedral

12 Sub-Shell Capacities & Quantum Numbers

Syllabus 1.3

- (a) State the maximum number of electrons that can occupy an *s*, a *p*, and a *d* sub-shell. [3]

- (b) Define the term *principal quantum number*. [1]

13 Sub-Shell Energy Order

Syllabus 1.3

- (a) List the following sub-shells in order of increasing energy: 3d, 2s, 4s, 1s, 3p, 2p, 3s. [2]

14 Sub-Shell Energy Level Diagram

Syllabus 1.3

The diagram below shows the relative energies of the sub-shells up to 3d.

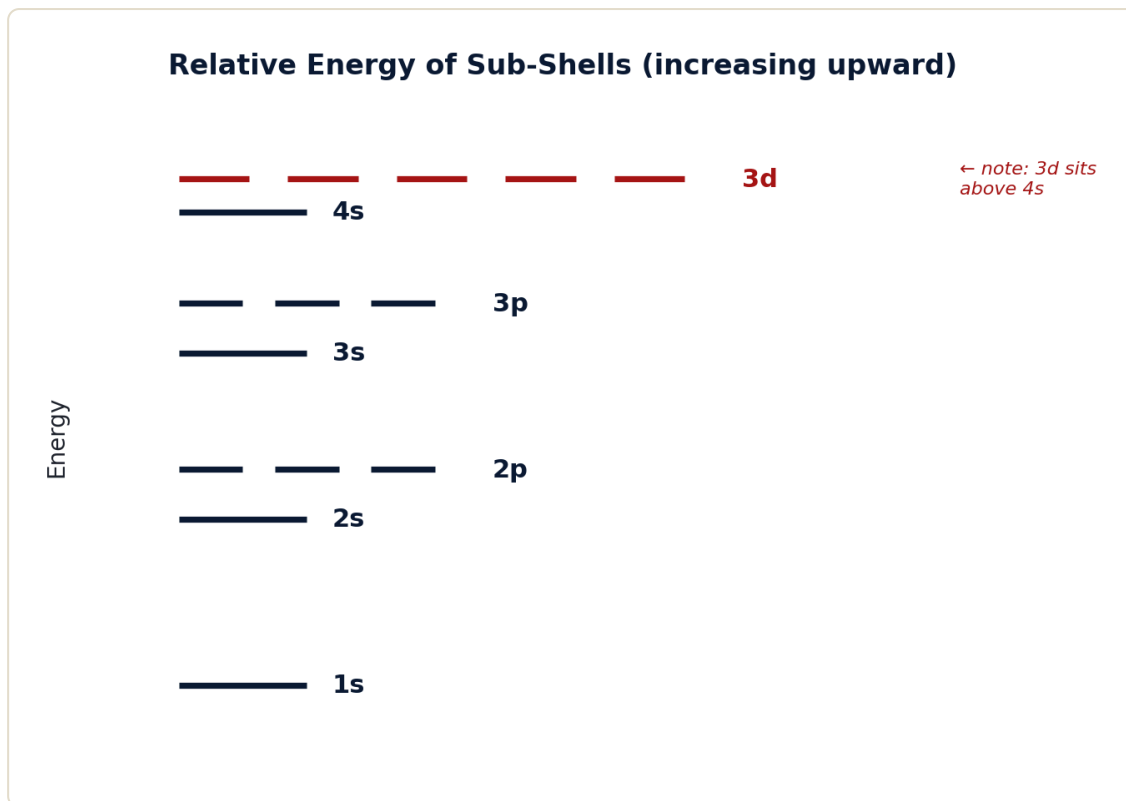


Fig. 14.1 Relative sub-shell energies

- (a) Identify, using the diagram, which sub-shell has a higher energy: 4s or 3d. [1]
-
- (b) Explain why this ordering means that the 4s sub-shell fills before the 3d sub-shell when building up the electron configuration of an atom. [2]
-
-
- (c) State why this same ordering means that the 4s electrons are removed **before** the 3d electrons when a transition metal atom forms a positive ion. [2]
-
-
- (d) Using the diagram, explain why the 2p sub-shell is drawn as three separate short lines rather than one. [1]
-

15 Orbital Shapes

Syllabus 1.3

- (a) Sketch the shape of a single *s* orbital and a single *p* orbital.

[2]

- (b) State one similarity and one difference between the three *p* orbitals in a given *p* sub-shell.

[2]

16 Orbitals in the $n = 3$ Shell

Syllabus 1.3

- (a) State the number of orbitals present in the $n = 3$ shell in total ($3s + 3p + 3d$), and hence the maximum number of electrons the $n = 3$ shell can hold.

[3]

THURSDAY · 20 AUG — TOPIC 1.3: ELECTRON CONFIGURATION & ORBITALS (PART 2)
17 Multiple Choice

Syllabus 1.3

- (a) The full electron configuration of a chlorine atom ($Z = 17$) is:

[1]

A $1s^2 2s^2 2p^6 3s^2 3p^6$

B $1s^2 2s^2 2p^6 3s^2 3p^5$

C $1s^2 2s^2 2p^6 3s^1 3p^6$

D $1s^2 2s^6 2p^2 3s^2 3p^5$

- (b) Which of these species contains an unpaired electron (is a free radical)?

[1]

A Cl^-

B $\text{Cl}^\bullet$

C Cl^{2-}

D HCl

- (c) Hund's rule states that electrons in orbitals of equal energy will:

[1]

A pair up as soon as possible

B occupy separate orbitals with parallel spin before pairing

C always remain unpaired

D occupy the lowest orbital only

- (d) The shorthand electron configuration of Fe^{3+} ($Z = 26$) is:

[1]

A $[\text{Ar}]3d^6$

B $[\text{Ar}]3d^5$

C $[\text{Ar}]3d^6 4s^1$

D $[\text{Ar}]3d^4 4s^1$
18 Full Electron Configurations

Syllabus 1.3

- (a) Write the full electron configuration of a chromium atom ($Z = 24$). Comment on anything unusual about it.

[2]

- (b) Write the full electron configuration of a copper atom ($Z = 29$). Comment on anything unusual about it.

[2]

19 Shorthand Configuration of Ions

Syllabus 1.3

- (a) Write the shorthand (noble-gas core) electron configuration of the Fe^{2+} ion.

[2]

.....

20 Electrons-in-Boxes

Syllabus 1.3

- (a) Draw an electrons-in-boxes diagram for the 2p sub-shell of a nitrogen atom ($Z = 7$).

[2]

.....

- (b) Draw an electrons-in-boxes diagram for the 3d sub-shell of a Cr^{3+} ion ($Z = 24$).

[2]

.....

21 Free Radicals

Syllabus 1.3

- (a) Define the term *free radical*.

[1]

.....

- (b) Of the three species $\text{Cl}^\bullet$, Cl^- and Cl_2 , identify which is a free radical and explain your choice.

[3]

.....

.....

.....

22 From Configuration to Identity

Syllabus 1.3 (synoptic)

- (a) An atom has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^4$. State the group and period of the Periodic Table this element belongs to, and identify the element.

[3]

.....

.....

.....

- (b) State the electron configuration of the $2-$ ion formed by this element.

[2]

.....

FRIDAY · 21 AUG — TOPIC P5: PAPER 5 FOCUS: PLANNING & ANALYSIS

23 Planning an Investigation

Paper 5 — Planning

A student wants to investigate how the concentration of hydrochloric acid affects the rate of its reaction with calcium carbonate (marble chips).

- (a) Identify the independent variable and the dependent variable in this investigation. [2]

.....

.....

- (b) State **two** variables that should be kept constant, and explain why each must be controlled. [4]

.....

.....

.....

- (c) Describe how the dependent variable could be measured, including the apparatus you would use. [2]

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.....

- (d) Suggest a suitable range and number of concentration values to use, and justify your choice. [2]

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24 Plotting & Analysing Results

Paper 5 — Analysis

One run of the investigation (using one particular acid concentration) produced the volume of CO₂ gas collected against time shown below.

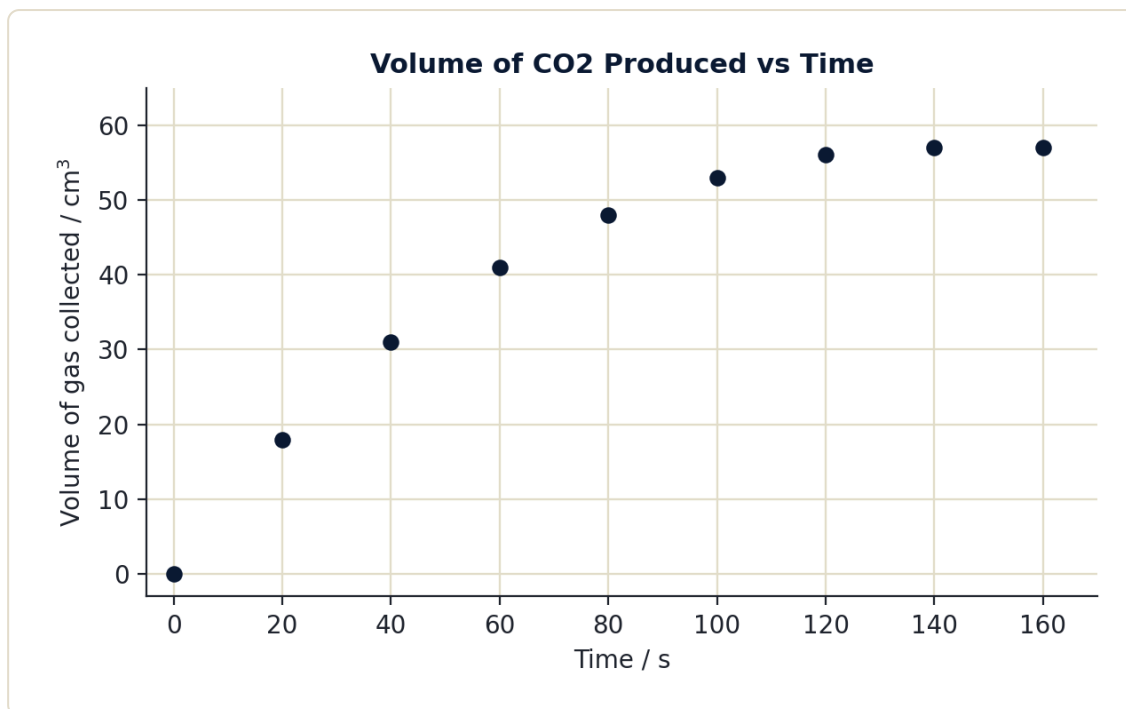


Fig. 24.1 Volume of gas produced vs time

- (a) Draw a smooth best-fit curve through the plotted points. [1]
- (b) By drawing a tangent to the curve at $t = 0$, determine the initial rate of reaction, in $\text{cm}^3 \text{s}^{-1}$. Show your working. [3]
- (c) State, with a reason, the time at which the reaction appears to have finished. [2]
- (d) Explain, in terms of collision theory, why the rate of gas production decreases over the course of the reaction. [2]

25 Evaluation

Paper 5 — Evaluation

- (a) The student repeated this run three times and obtained slightly different curves each time. Suggest **two** sources of error that could explain this variation. **[4]**

- (b) For **one** of the errors you identified in (a), suggest a specific practical improvement to reduce its effect. **[2]**

- (c) Comment on the reliability of using only one run of data to draw a conclusion, and suggest how the student could improve the reliability of their conclusion. **[2]**

MIXED PRACTICE · FULL WEEK — TOPIC 1.1–1.4: FULL-WEEK SYNTHESIS QUESTION

26 Element Y — A Synoptic Data-Response Question

Syllabus 1.1–1.4

The graph below shows the successive ionisation energies of Element Y, an element in Period 3.

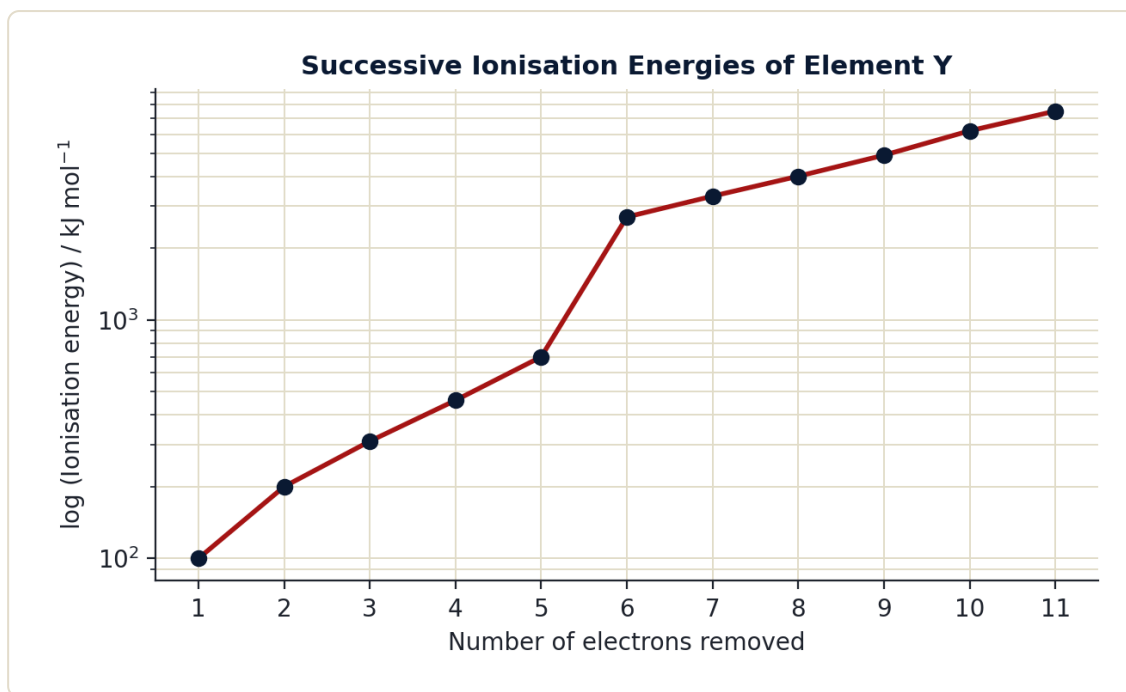


Fig. 26.1 Successive ionisation energies of Element Y (log scale)

- (a) Use the graph to deduce the group of the Periodic Table Element Y belongs to. Explain your reasoning fully. [3]
-
-
-
- (b) Suggest the full electron configuration of Element Y. [2]
-
-
-
- (c) State and explain how the first ionisation energy would change on moving from Element Y to the next element in the same period. [3]
-
-
-

26 Element Y (continued)

Syllabus 1.1–1.4

- (d) Element Y has two naturally occurring isotopes. Explain what is meant by the term *isotope*, and explain why both isotopes of Y have identical chemical properties. [3]

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.....

- (e) A sample of Element Y is analysed by mass spectrometry. Explain how the m/z value of the atomic ion peak of highest mass relates to the relative isotopic mass of Y's heavier isotope. [2]

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.....

26 Element Y — Extended Writing

Syllabus 1.1–1.4

- (f) "Ionisation energy always increases across a period." Evaluate this statement, using ideas about atomic structure and any relevant exceptions you know of within Period 2 or Period 3. [7]

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.....

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MARK SCHEME — WEEK 1

Q	MARKING POINTS	MARKS
1(a)	C — proton	1
1(b)	B — 16	1
1(c)	A — the positive plate (electrons are negatively charged)	1
1(d)	A — radius decreases, ionisation energy increases	1
1(e)	C — Na^+ (loses its entire outer shell)	1
1(f)	B — the number of occupied electron shells increases	1

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
2	Proton: charge +1, mass 1. Neutron: charge 0, mass 1. Electron: charge -1, mass $1/1840$ (or negligible). 1 mark per correct row.	3
3(a)	Protons are deflected towards the negative plate (1), because they carry a positive charge and are attracted to it (1).	2
3(b)	Neutrons carry no charge, so experience no electrostatic force from the field.	1
4(a)	Silicon ≈ 117 pm; Chlorine ≈ 99 pm (accept ± 3 pm from the plotted line).	2
4(b)	Atomic radius decreases across the period.	1
4(c)	Nuclear charge increases across the period (1); while the number of shielding shells stays the same (1); so the increased attraction between nucleus and outer electrons pulls them closer, decreasing radius (1).	3

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
5(a)	Na^+ has one fewer electron shell than Na (the outer 3s electron is lost) (1); with the same nuclear charge now attracting fewer electrons, pulling the remaining shells in closer (1).	2
5(b)	Cl^- has one more electron than Cl in the same outer shell (1); the same nuclear charge now attracts more electrons, and increased electron-electron repulsion pushes the shell outward (1).	2
6(a)	B — the same number of protons, different neutrons	1
6(b)	C — density of the pure element	1
6(c)	B — mass-to-charge ratio (m/z)	1
6(d)	B — 20.2 ($0.90 \times 20 + 0.10 \times 22 = 20.2$)	1

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
7(a)	Atoms of the same element (same proton/atomic number) (1) with different numbers of neutrons (different nucleon/mass number) (1).	2
7(b)	$^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$ (nucleon number top-left, proton number bottom-left). 1 mark each.	2
8(a)	Both isotopes have the same number of protons and electrons (1), and therefore the same electron configuration, which determines chemical behaviour (1).	2
8(b)	Any one: density / rate of diffusion / precise melting or boiling point (1); explained by the difference in mass between the isotopes (1).	2
9(a)	Three isotopes.	1
9(b)	$\text{Ar} = (28 \times 92.2 + 29 \times 4.7 + 30 \times 3.1) \div 100$ (1); $= (2581.6 + 136.3 + 93) \div 100$ (1); $= 28.1$ (1).	3

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
9(c)	This isotope (mass 29) is much less abundant / naturally occurs in a much smaller proportion than the mass-28 isotope.	1
10(a)	Any sensible application with correct linking property, e.g. carbon-14 dating (radioactive isotope decays at a known rate) / medical tracers (radioactive isotopes emit detectable radiation while behaving chemically like the stable isotope) / isotopic labelling in reaction mechanism studies.	3
11(a)	B — 6	1
11(b)	C — 5	1
11(c)	C — 4s	1
11(d)	B — spherical	1

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
12(a)	s = 2, p = 6, d = 10.	3
12(b)	The quantum number (<i>n</i>) that indicates the main energy level / shell an electron occupies.	1
13(a)	1s, 2s, 2p, 3s, 3p, 4s, 3d (full marks for this order; 1 mark if only the 4s/3d position is wrong).	2
14(a)	4s has a lower energy than 3d (i.e. 3d is higher).	1
14(b)	Electrons fill the lowest-energy sub-shells available first (Aufbau principle) (1); since 4s is lower in energy than 3d, it is filled first (1).	2
14(c)	Once filled, the relative energies of 4s and 3d effectively swap / 3d becomes lower in energy than 4s for the ion (1); so the higher-energy 4s electrons are removed first (1).	2

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
14(d)	The 2p sub-shell consists of three orbitals of equal energy (degenerate), shown as three separate lines at the same height.	1
15(a)	s orbital: a sphere. p orbital: a dumb-bell / figure-of-eight shape with two lobes.	2
15(b)	Similarity: all three have the same shape/energy (are degenerate). Difference: each points along a different axis (x, y, z).	2
16(a)	9 orbitals in total (1 s + 3 p + 5 d) (1); maximum of 18 electrons (1).	3
17(a)	B	1
17(b)	B — Cl•	1

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
17(c)	B — occupy separate orbitals with parallel spin before pairing	1
17(d)	B — $[Ar]3d^5$	1
18(a)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ (1); unusual because a single 4s electron is present rather than the expected $4s^2$ — a half-filled 3d sub-shell is extra stable (1).	2
18(b)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ (1); unusual because a single 4s electron is present — a fully-filled 3d sub-shell is extra stable (1).	2
19(a)	$[Ar]3d^6$ (1 mark for removing the two 4s electrons first, 1 mark for the correct remaining configuration).	2
20(a)	Three boxes for the three 2p orbitals; two contain one arrow each pointing the same way, one contains one arrow (Hund's rule — N has 3 electrons in 2p, one per orbital before pairing).	2

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
20(b)	Three boxes for the 3d orbitals; three of the five boxes contain one arrow each pointing the same way (Cr^{3+} has $3d^3$, giving 3 unpaired electrons across separate orbitals).	2
21(a)	A species (atom, ion or molecule) with an unpaired electron.	1
21(b)	$Cl\cdot$ is the free radical (1); because Cl^- has a full outer shell with all electrons paired (1), and Cl_2 has all electrons paired within the covalent bond / lone pairs (1).	3
22(a)	Group 17 (1); Period 3 (1); chlorine (1) — deduced from 7 electrons in the outer ($n=3$) shell.	3
22(b)	$1s^2 2s^2 2p^6 3s^2 3p^6$ (i.e. same as argon — gains one electron to fill the 3p sub-shell).	2
23(a)	Independent variable: concentration of hydrochloric acid. Dependent variable: rate of reaction, e.g. volume of gas produced per unit time.	2

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
23(b)	Any two of, each with a correct explanation: mass/size of marble chips kept constant (so surface area does not affect rate); temperature kept constant, e.g. using a water bath (since temperature also affects rate); volume of acid used kept constant (so total moles of acid available is comparable). 2 marks per correctly explained variable.	4
23(c)	Collect the gas produced using a gas syringe (or by downward displacement of water) and record the volume of gas at fixed time intervals, or record the time taken to collect a fixed volume of gas.	2
23(d)	A suitable range, e.g. 0.5, 1.0, 1.5, 2.0, 2.5 mol dm ⁻³ (at least 5 values), justified as giving a wide enough spread to show a clear trend without being too dilute to react in a measurable time or too concentrated to control safely.	2
24(a)	A smooth curve of decreasing gradient through the points, levelling off/plateauing towards the later time readings.	1
24(b)	Tangent drawn at t = 0 (1); gradient calculated correctly from the tangent using a sensible triangle (1); correct value with unit cm ³ s ⁻¹ (accept a range around 1.0–1.1 cm ³ s ⁻¹ depending on the tangent drawn) (1).	3
24(c)	At approximately t = 140–160 s (1); because the volume of gas has stopped increasing / the curve has flattened, showing the reaction has stopped (1).	2

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
24(d)	As the reaction proceeds, the concentration of HCl decreases (1); this reduces the frequency of collisions between reactant particles, so the rate of gas production decreases (1).	2
25(a)	Any two sensible sources of error, e.g. inconsistent marble chip size/surface area between repeats; small variation in acid volume measured out; gas escaping before the bung/seal was fully in place; timing reaction errors (human reaction time). 2 marks each for a clearly explained source.	4
25(b)	A specific, linked improvement, e.g. use a fixed mass of powdered/uniform-sized marble chips rather than irregular chips, to keep surface area consistent between repeats.	2
25(c)	A single run cannot show whether a result is reliable/repeatable (1); the student should repeat the experiment at each concentration and calculate a mean, discarding any clear anomalies (1).	2
26(a)	Group 15 (1); because there is a large jump in ionisation energy between the 5th and 6th electrons removed (1), showing 5 electrons in the outer shell before a much more strongly held inner-shell electron is reached (1).	3
26(b)	1s ² 2s ² 2p ⁶ 3s ² 3p ³ (1); consistent with Period 3, Group 15, e.g. phosphorus (1).	2

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
26(c)	First ionisation energy increases (1); because nuclear charge increases by one proton while shielding stays approximately the same, so the outer electron is held more strongly and more energy is needed to remove it (1); (accept mention of a relevant Group 16 anomaly if discussed) (1).	3
26(d)	Isotope: atoms of the same element (same proton number) with different numbers of neutrons (different nucleon number) (1). Both isotopes have identical chemical properties because they have the same number of protons and electrons, and therefore the same electron configuration (1), which determines chemical bonding behaviour (1).	3
26(e)	The m/z of the highest-mass ion peak corresponds to the mass of the heaviest isotope present (1), since m/z for a singly-charged ion is numerically equal to its isotopic mass (1).	2
26(f)	Level-of-response question (7 marks) — see indicative content and mark bands below.	7

QUESTION 26(F) — INDICATIVE CONTENT

- ▶ The general trend across a period is that first ionisation energy increases, because nuclear charge increases while shielding stays roughly constant, so outer electrons are held more tightly.
- ▶ However, this trend is not perfectly smooth: there is a small dip/anomaly between Group 13 and Group 14 (e.g. Al lower than Mg), because the outer electron in Group 13 occupies a p sub-shell that is slightly higher in energy and further from the nucleus than the filled s sub-shell in Group 12/2.
- ▶ There is a second dip/anomaly between Group 15 and Group 16 (e.g. S lower than P), because in Group 16 two electrons must now pair up within one p orbital, and the extra repulsion between the paired electrons makes one easier to remove than expected.
- ▶ A full evaluation should state the general trend AND at least one specific exception, explaining the electronic reason behind the exception, rather than treating the statement as simply true or simply false.
- ▶ The strongest answers use specific Period 2 or Period 3 examples (e.g. Mg to Al, or P to S) to support the explanation.

BAND	DESCRIPTOR
Level 1 (1–3 marks)	States the general increasing trend OR identifies an exception, with limited or no electronic explanation.
Level 2 (4–5 marks)	States the general trend and at least one exception, with a partially correct electronic explanation.
Level 3 (6–7 marks)	A detailed, accurate evaluation stating the general trend, at least one specific exception with a named example, and a fully correct electronic explanation for why the exception occurs.

A NOTE ON SOURCING REAL PAST PAPERS

- These questions are original, written to match Cambridge 9701's exact command words, mark allocations and graph/data-response style for Topic 1 — they are not reproductions of real past exam papers, which are copyrighted by Cambridge Assessment.
- For genuine past-paper practice on this week's content, use the following real Save My Exams topic-question pages ([savemyexams.com](https://www.savemyexams.com) > A Level > Chemistry > CIE): **Atomic Structure — Particles in the Atom & Atomic Radius**; **Atomic Structure — Isotopes**; **Atomic Structure — Electrons, Energy Levels & Atomic Orbitals**; **Atomic Structure — Ionisation Energy**.
- For Friday's Paper 5 skills, use real Cambridge 9701 Paper 5 past papers and mark schemes (any recent June/November series) from the Cambridge School Support Hub.